

Grade 11 Multi-Subject Practice Examination

Mathematics - Chemistry - Physics - Biology - English

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Name: _____ Class: _____ Date: _____

Instructions

This paper contains five subjects, each with a mix of question formats: choose-the-answer questions, true/false statements, blank-completion sentences, tables to complete, a classification table, short-response questions, and longer written responses.

Read every question carefully. Write your answers in the space provided. For longer written responses, use the lined space given and continue on extra paper if needed.

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Chemistry

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Mathematics

Answer all questions in the spaces provided.

Choose the letter of the correct answer for each question.

1. $\log_2(8)$ is equal to:
(A) 2
(B) 3
(C) 4
(D) 8
2. Simplify: $3^4 \times 3^2$
(A) 3^6
(B) 3^8
(C) 9^6
(D) 3^2
3. If $\sin(\theta) = 3/5$ and θ is an acute angle, what is $\cos(\theta)$?
(A) $4/5$
(B) $3/4$
(C) $5/3$
(D) $5/4$
4. What is the next term in the arithmetic sequence 2, 5, 8, 11, ...?
(A) 13
(B) 14
(C) 15
(D) 16
5. Find the sum of the first 5 terms of the geometric sequence 2, 6, 18, 54, ...
(A) 121
(B) 242
(C) 243
(D) 244
6. What is the mean of the data set {4, 8, 6, 10, 12}?
(A) 6
(B) 7
(C) 8
(D) 9
7. A fair coin is tossed twice. What is the probability of getting exactly one head?
(A) $1/4$
(B) $1/2$
(C) $3/4$
(D) 1
8. $\log_{10}(100)$ is equal to:
(A) 1
(B) 2
(C) 10
(D) 100

9. Solve for x : $\log_3(x) = 4$.
- (A) 12
 - (B) 64
 - (C) 81
 - (D) 3
10. The value of $\ln(e)$ (the natural logarithm of e) is:
- (A) 0
 - (B) 1
 - (C) e
 - (D) 10
11. Which of the following is a term of the geometric sequence with $a_1 = 3$ and $r = 2$?
- (A) 8
 - (B) 12
 - (C) 14
 - (D) 20
12. The variance of a data set is the square of its:
- (A) mean
 - (B) median
 - (C) mode
 - (D) standard deviation
13. How many different ways can 3 distinct books be arranged in a row on a shelf?
- (A) 3
 - (B) 6
 - (C) 9
 - (D) 12
14. If $P(A) = 0.3$ and $P(B) = 0.5$, and events A and B are mutually exclusive, what is $P(A \text{ or } B)$?
- (A) 0.15
 - (B) 0.2
 - (C) 0.8
 - (D) 1.0
15. Write 3^{-2} as a fraction.
- (A) -9
 - (B) $-1/9$
 - (C) $1/9$
 - (D) 9
16. If $a_n = 2n + 1$, what is the 6th term of the sequence?
- (A) 11
 - (B) 12
 - (C) 13
 - (D) 14
17. $2 \log(5)$ is equal to:
- (A) $\log(10)$
 - (B) $\log(25)$
 - (C) $\log(7)$
 - (D) $\log(2.5)$

18. The sum of the interior angles of a quadrilateral is:
- (A) 180
 - (B) 270
 - (C) 360
 - (D) 450
19. A fair six-sided die is rolled once. What is $P(\text{rolling an even number})$?
- (A) $\frac{1}{6}$
 - (B) $\frac{1}{3}$
 - (C) $\frac{1}{2}$
 - (D) $\frac{2}{3}$
20. Simplify $(2^3)^2$.
- (A) 2^5
 - (B) 2^6
 - (C) 2^8
 - (D) 2^9
21. An arithmetic sequence has first term 7 and common difference -2. What is the 4th term?
- (A) -1
 - (B) 0
 - (C) 1
 - (D) 2
22. In a class, test scores of 70 occur 8 times, 80 occurs 5 times, and 90 occurs 3 times. What is the mode?
- (A) 70
 - (B) 80
 - (C) 90
 - (D) Cannot be determined
23. What is the square root of 100?
- (A) 10
 - (B) 50
 - (C) 100
 - (D) 1000
24. A standard deck of 52 playing cards has 26 red cards. What is the probability of drawing a red card?
- (A) $\frac{1}{4}$
 - (B) $\frac{1}{2}$
 - (C) $\frac{1}{13}$
 - (D) $\frac{1}{26}$
25. Which pair(s) of angles are complementary (sum to 90 degrees)?
- (A) 30 and 60 degrees
 - (B) 45 and 45 degrees
 - (C) Both of the above
 - (D) 100 and 80 degrees
26. If the mode of a data set is 15, this means that:
- (A) 15 is the largest value
 - (B) 15 occurs most frequently
 - (C) 15 is the average of the data
 - (D) 15 is the middle value

Circle T if the statement is true, or F if it is false.

27. T / F $\log(ab) = \log(a) + \log(b)$, for $a, b > 0$.
28. T / F $\sin(90 \text{ degrees}) = 0$.
29. T / F An arithmetic sequence has a constant common ratio between consecutive terms.
30. T / F Standard deviation is always a non-negative number.
31. T / F The probability of any event always lies between 0 and 1, inclusive.
32. T / F $\cos(0 \text{ degrees}) = 1$.
33. T / F In the geometric sequence 3, 6, 12, 24, the common ratio is 3.
34. T / F The median of a data set is always equal to its mean.
35. T / F The value of $\log(1)$ is 0 (for any valid base).
36. T / F Every arithmetic sequence is also a geometric sequence.
37. T / F Standard deviation measures how spread out data values are around the mean.
38. T / F The probability of two mutually exclusive events happening at the same time is always 0.
39. T / F $4!$ (4 factorial) is equal to 24.
40. T / F A sample space lists all the possible outcomes of a random experiment.
41. T / F The mode of a data set is always a single, unique value.
42. T / F $\log(0)$ is defined and equal to 0.
43. T / F The sum of the exterior angles of any convex polygon is always 360 degrees.
44. T / F A geometric sequence can have a negative common ratio.
45. T / F The range of a data set is the difference between its highest and lowest values.
46. T / F Combinations count the order of selection, while permutations do not.
47. T / F The probability of the complement of event A is $1 - P(A)$.
48. T / F $\log(a/b) = \log(a) - \log(b)$, for $a, b > 0$.
49. T / F The square root of 100 is 10.
50. T / F A standard deck of playing cards contains 52 cards, half of which are red.
51. T / F Two angles that sum to 90 degrees are called supplementary angles.
52. T / F The mode of a data set refers to the value that occurs most frequently.

Fill in each blank with the correct word, phrase, or value.

53. $\log_b(b) = \underline{\hspace{2cm}}$
54. The n th term of an arithmetic sequence is given by $a_n = a_1 + (n-1) \times \underline{\hspace{2cm}}$.
55. $\sin^2(\theta) + \cos^2(\theta) = \underline{\hspace{2cm}}$
56. The sum of the interior angles of a triangle is $\underline{\hspace{2cm}}$ degrees.
57. A sequence in which each term is found by multiplying the previous term by a fixed number is called $a(n)$ $\underline{\hspace{2cm}}$ sequence.
58. The middle value of an ordered data set is called the $\underline{\hspace{2cm}}$.

59. $5^0 =$ ____
60. The probability of an impossible event is ____.
61. Any nonzero number raised to the power 0 is equal to ____.
62. $n!$ (n factorial) is the product of all positive integers from 1 up to ____.
63. The average of the squared deviations of data values from the mean is called the ____.
64. $\log(1) =$ ____ (for any valid base).
65. The sum of the first n terms of an arithmetic sequence is $S_n = (n/2)(2a_1 + (n-1)\text{____})$.
66. The set of all possible outcomes of a random experiment is called the ____.
67. Events that cannot both happen at the same time are called ____ events.
68. The square root of the variance is called the ____.
69. The sum of the exterior angles of any convex polygon is always ____ degrees.
70. $\log(a/b) = \log(a) - \log(\text{____})$.
71. An ordered arrangement of objects, where order matters, is called $a(n)$ ____.
72. A selection of objects where order does not matter is called $a(n)$ ____.
73. The difference between the highest and lowest values in a data set is called the ____.
74. $P(\text{not } A) = 1 -$ ____.
75. Two angles that sum to 90 degrees are called ____ angles.
76. Two angles that sum to 180 degrees are called ____ angles.
77. The square root of a perfect square such as 100 is a whole, ____ number.
78. A standard deck of playing cards contains ____ cards in total.

79. Table 1 - Standard trigonometric values (decimals rounded to 2 d.p. where needed)

Angle	sin	cos	tan
0 degrees	0.00	1.00	0.00
30 degrees	0.50	_____	0.58
45 degrees	_____	0.71	1.00
60 degrees	0.87	0.50	_____
90 degrees	1.00	_____	undefined

80. Table 2 - Frequency table of exam scores (25 students)

Score range	Frequency	Cumulative frequency
50-59	4	4
60-69	_____	10
70-79	8	_____
80-89	5	23

90-99	_____	25
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81. Table 3 - Converting between exponential and logarithmic form

Exponential form	Logarithmic form
$2^3 = 8$	_____
$10^2 = 100$	_____
$5^0 = 1$	_____

82. Classify each sequence below by putting a tick in the correct column.

Item	Arithmetic	Geometric	Neither
2, 4, 6, 8, 10, ...			
3, 6, 12, 24, ...			
1, 4, 9, 16, ...			
100, 90, 80, 70, ...			
5, 15, 45, 135, ...			
2, 3, 5, 7, 11, ...			
-8, -4, 0, 4, ...			
1, -2, 4, -8, ...			

Answer in one to three complete sentences.

83. Solve for x: $2^x = 32$. Show your working.

84. Find the 10th term of the arithmetic sequence with $a_1 = 4$ and common difference $d = 3$.

85. A die is rolled once. What is the probability of rolling a number greater than 4?

86. Convert the logarithmic equation $\log_5(25) = 2$ into exponential form.

87. Find the mean and the range of the data set: 12, 15, 9, 18, 21.

88. Calculate $4!$ (4 factorial).

89. A bag contains 3 red balls and 2 blue balls. One ball is drawn at random. What is the probability that it is red?

92. Explain, using an example of each, the difference between an arithmetic sequence and a geometric sequence. Then describe one real-world situation that could be modelled by each type of sequence.

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93. Explain what standard deviation measures, and describe what it means for a data set of test scores to have a small standard deviation compared with a large standard deviation.

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Chemistry

Answer all questions in the spaces provided.

Choose the letter of the correct answer for each question.

- Which of the following does NOT affect the rate of a chemical reaction?
 - Temperature
 - Concentration
 - A catalyst
 - The color of the container
- A catalyst speeds up a reaction by:
 - Increasing the temperature
 - Lowering the activation energy
 - Increasing the concentration
 - Increasing the pressure
- According to Le Chatelier's principle, increasing the pressure on a gaseous equilibrium system shifts the equilibrium toward the side with:
 - Fewer gas molecules
 - More gas molecules
 - No change occurs
 - Cannot be determined
- A solution with pH = 3 is:
 - Strongly basic
 - Weakly basic
 - Neutral
 - Acidic
- Which of the following is classified as a strong acid?
 - Acetic acid (CH_3COOH)
 - Hydrochloric acid (HCl)
 - Ammonia (NH_3)
 - Carbonic acid (H_2CO_3)
- In a redox reaction, oxidation is defined as:
 - Gain of electrons
 - Loss of electrons
 - Gain of protons
 - Loss of protons
- How many moles are in 44 g of CO_2 ? (molar mass of CO_2 = 44 g/mol)
 - 0.5
 - 1
 - 2
 - 4
- At chemical equilibrium:
 - The reaction has completely stopped
 - The forward and reverse reaction rates are equal
 - Only the forward reaction occurs
 - Reactant and product concentrations are always equal

9. The molar mass of water (H_2O) is approximately:
- (A) 16 g/mol
 - (B) 18 g/mol
 - (C) 20 g/mol
 - (D) 44 g/mol
10. Which gas is typically produced when a reactive metal reacts with a dilute acid?
- (A) Oxygen
 - (B) Hydrogen
 - (C) Nitrogen
 - (D) Carbon dioxide
11. The pH scale commonly used in chemistry ranges from:
- (A) 0 to 7
 - (B) 0 to 10
 - (C) 0 to 14
 - (D) 1 to 100
12. An exothermic reaction:
- (A) Absorbs heat from the surroundings
 - (B) Releases heat to the surroundings
 - (C) Has no energy change
 - (D) Only occurs in solids
13. Which subatomic particle carries a negative electric charge?
- (A) Proton
 - (B) Neutron
 - (C) Electron
 - (D) Nucleus
14. The process of a gas changing directly into a solid, without becoming a liquid, is called:
- (A) Evaporation
 - (B) Sublimation
 - (C) Deposition
 - (D) Condensation
15. Which statement best describes an isotope?
- (A) Atoms with a different number of protons
 - (B) Atoms of the same element with different numbers of neutrons
 - (C) Atoms with different numbers of electrons only
 - (D) Molecules that share the same formula
16. Which type of bond involves atoms sharing pairs of electrons?
- (A) Ionic bond
 - (B) Covalent bond
 - (C) Metallic bond
 - (D) Hydrogen bond
17. The chemical formula for sulfuric acid is:
- (A) H_2SO_3
 - (B) H_2SO_4
 - (C) HSO_4
 - (D) SO_4

18. Which of the following is an example of a physical change?
- (A) Burning paper
 - (B) Melting wax
 - (C) Rusting of a nail
 - (D) Baking a cake
19. How many valence electrons does a Group 1 (alkali metal) element typically have?
- (A) 1
 - (B) 2
 - (C) 7
 - (D) 8
20. A pure substance made of only one type of atom is called a(n):
- (A) Compound
 - (B) Mixture
 - (C) Element
 - (D) Solution
21. The technique used to separate a mixture based on differences in boiling point is called:
- (A) Filtration
 - (B) Distillation
 - (C) Chromatography
 - (D) Decantation
22. A solution that has dissolved as much solute as it possibly can at a given temperature is described as:
- (A) Dilute
 - (B) Concentrated
 - (C) Saturated
 - (D) Supersaturated
23. The chemical symbol for potassium is:
- (A) K
 - (B) P
 - (C) Po
 - (D) Pt
24. Which of these elements naturally exists as a two-atom (diatomic) molecule?
- (A) Helium
 - (B) Neon
 - (C) Oxygen
 - (D) Argon
25. The process of a solid changing directly into a gas, without passing through the liquid state, is called:
- (A) Melting
 - (B) Sublimation
 - (C) Evaporation
 - (D) Condensation
26. Which subatomic particle carries no electric charge?
- (A) Proton
 - (B) Electron
 - (C) Neutron
 - (D) Ion

Circle T if the statement is true, or F if it is false.

27. T / F Increasing the temperature of a reaction generally increases its rate.
28. T / F A catalyst is consumed during the reaction that it speeds up.
29. T / F A pH of 7 indicates a neutral solution at 25 degrees C.
30. T / F Adding more reactant to a system at equilibrium shifts the equilibrium toward the products.
31. T / F Bases turn red litmus paper blue.
32. T / F In electrolysis, reduction always occurs at the anode.
33. T / F Increasing the surface area of a solid reactant generally has no effect on reaction rate.
34. T / F A solution with pH 1 is more acidic than a solution with pH 5.
35. T / F Exothermic reactions release energy to the surroundings, usually as heat.
36. T / F All acids contain the element oxygen.
37. T / F Ionic bonds form through the transfer of electrons between atoms.
38. T / F Endothermic reactions release heat to the surroundings.
39. T / F Isotopes of an element have the same number of protons but different numbers of neutrons.
40. T / F A covalent bond involves the complete transfer of an electron from one atom to another.
41. T / F The pH of pure water at 25 degrees C is 7.
42. T / F Metal atoms typically lose electrons to form positively charged ions.
43. T / F A mixture can be separated by physical means, while a compound cannot be separated without a chemical reaction.
44. T / F Noble gases are generally very reactive elements.
45. T / F A concentrated solution contains a relatively large amount of dissolved solute.
46. T / F Filtration can be used to separate a dissolved (soluble) salt from water.
47. T / F The atomic number of an element equals the number of protons in its nucleus.
48. T / F All solutions are mixtures, but not all mixtures are solutions.
49. T / F The chemical symbol for potassium is K, from its Latin name "kalium."
50. T / F Oxygen naturally exists as a diatomic molecule, O₂.
51. T / F Sublimation is the change of state from solid directly to gas.
52. T / F Neutrons carry a positive electric charge.

Fill in each blank with the correct word, phrase, or value.

53. The branch of chemistry that studies the speed of reactions is called chemical ____.
54. A substance that increases the rate of a reaction without being permanently consumed is a(n) ____.
55. At equilibrium, the rate of the forward reaction equals the rate of the ____ reaction.
56. A solution with a pH less than 7 is described as ____.
57. In the Bronsted-Lowry theory, a species that donates a proton (H⁺) is called a(n) ____.
58. In an electrochemical cell, oxidation occurs at the electrode called the ____.
59. One mole of any substance contains approximately ____ particles.

60. The reaction between an acid and a base that produces a salt and water is called a(n) ____ reaction.
61. A reaction that absorbs heat energy from its surroundings is called ____.
62. The smallest particle of an element that retains its chemical properties is called a(n) ____.
63. Atoms that have gained or lost electrons, giving them an electric charge, are called ____.
64. The number of protons in the nucleus of an atom is called its ____ number.
65. A bond formed by the transfer of electrons from a metal to a non-metal is called a(n) ____ bond.
66. The substances present at the start of a chemical reaction are called the ____.
67. The horizontal rows of the periodic table are called ____.
68. The vertical columns of the periodic table are called ____ (or families).
69. A pure substance made of only one type of atom is called a(n) ____.
70. A pure substance made of two or more elements chemically combined in fixed proportions is called a(n) ____.
71. The technique used to separate a mixture based on differences in boiling point is called ____.
72. A solution containing the maximum amount of dissolved solute at a given temperature is described as ____.
73. Elements in the same vertical column of the periodic table are called a ____, and share similar chemical properties.
74. Elements in Group 18 of the periodic table are called the ____ gases, and are generally unreactive.
75. The chemical symbol for potassium is ____.
76. A change of state from solid directly to gas, without becoming a liquid, is called ____.
77. The subatomic particle with no electric charge is the ____.
78. Oxygen naturally exists as a diatomic molecule with the formula ____.

79. Table 1 - pH and the nature of a solution

pH value	Nature of solution	[H ⁺] compared with [OH ⁻]
2	acidic	_____
7	_____	[H ⁺] = [OH ⁻]
11	basic	_____

80. Table 2 - Mole calculations (moles = mass / molar mass)

Substance	Molar mass (g/mol)	Mass given (g)	Moles
NaCl	58.5	117	_____
H ₂ O	18	_____	2
CO ₂	44	22	_____

81. Table 3 - Concentration calculations (concentration = moles of solute / volume of solution in litres)

Moles of solute (mol)	Volume (L)	Concentration (mol/L)
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0.5	1	_____
_____	0.5	4
1.5	_____	0.5

82. For each change below, put a tick in the column that best describes it.

Item	Physical change	Chemical change
Melting ice		
Burning wood		
Rusting of iron		
Boiling water		
Digesting food		
Dissolving sugar in water		
Photosynthesis in a leaf		
Cutting a sheet of paper		

Answer in one to three complete sentences.

83. State two factors (other than a catalyst) that increase the rate of a chemical reaction.

84. Calculate the number of moles in 9 g of water. (molar mass of H₂O = 18 g/mol)

85. Write the word equation for the neutralization reaction between hydrochloric acid and sodium hydroxide.

86. Explain why increasing pressure shifts a gaseous equilibrium toward the side with fewer moles of gas.

87. A solution has a higher hydrogen ion concentration than a neutral solution. State whether it is acidic or basic, and give its approximate pH range.

88. Write the balanced chemical equation for the reaction between magnesium metal and hydrochloric acid, producing magnesium chloride and hydrogen gas.

89. Dissolving ammonium nitrate in water cools the resulting solution. State whether this process is exothermic or endothermic, and explain your reasoning.

90. An atom has 11 protons and 12 neutrons. State its atomic number and its mass number.

Write a well-organized, extended response of about 120-200 words.

91. Using collision theory, explain why increasing the temperature of a reaction increases its rate. Your answer should refer to particle motion, collision frequency, and the number of successful collisions.

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This image shows a full page of blank, lined paper. It features approximately 20 evenly spaced horizontal grey lines across its entire width, providing a template for writing or drawing. The margins are consistent on all sides.

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Physics

Answer all questions in the spaces provided.

Choose the letter of the correct answer for each question.

1. Ohm's Law states that:

- (A) $V = I/R$
- (B) $V = IR$
- (C) $V = R/I$
- (D) $I = VR$

2. The SI unit of electric current is the:

- (A) Volt
- (B) Ohm
- (C) Ampere
- (D) Watt

3. Two resistors of 4 ohm and 6 ohm are connected in series. What is the total resistance?

- (A) 2 ohm
- (B) 10 ohm
- (C) 24 ohm
- (D) 1.5 ohm

4. In a parallel circuit, the voltage across each branch is:

- (A) Different for each branch
- (B) The same across each branch
- (C) Always zero
- (D) Equal to the total current

5. A converging (convex) lens forms a real, inverted image when the object is placed:

- (A) Inside the focal length
- (B) Exactly at the focal point
- (C) Beyond the focal length
- (D) Only at infinity

6. Heat transfer through direct contact between vibrating particles is called:

- (A) Convection
- (B) Radiation
- (C) Conduction
- (D) Evaporation

7. The SI unit of electrical power is the:

- (A) Ampere
- (B) Volt
- (C) Watt
- (D) Coulomb

8. Which of these materials is the best electrical conductor?

- (A) Rubber
- (B) Glass
- (C) Copper
- (D) Wood

9. Specific heat capacity is defined as the amount of heat needed to raise the temperature of 1 kg of a substance by:
- (A) 1 degree Celsius
 - (B) 10 degrees Celsius
 - (C) 100 degrees Celsius
 - (D) 0.5 degree Celsius
10. A convex (diverging) mirror always produces a(n):
- (A) real, magnified image
 - (B) virtual, diminished image
 - (C) virtual, magnified image
 - (D) real, diminished image
11. Two 10 ohm resistors are connected in parallel. What is their combined resistance?
- (A) 5 ohm
 - (B) 10 ohm
 - (C) 20 ohm
 - (D) 50 ohm
12. Which of the following quantities is measured in joules?
- (A) Power
 - (B) Current
 - (C) Energy
 - (D) Resistance
13. The process of using a changing magnetic field to generate an electric current is called:
- (A) Magnetization
 - (B) Electromagnetic induction
 - (C) Conduction
 - (D) Polarization
14. A transformer that increases voltage (while decreasing current) is called a:
- (A) Step-down transformer
 - (B) Step-up transformer
 - (C) Isolation transformer
 - (D) Rectifier
15. Which method of heat transfer can occur through the vacuum of space?
- (A) Conduction only
 - (B) Convection only
 - (C) Radiation only
 - (D) All three equally
16. Which formula correctly gives the electrical energy transferred, E , in terms of power P and time t ?
- (A) $E = P/t$
 - (B) $E = P \times t$
 - (C) $E = P + t$
 - (D) $E = P - t$
17. The formula for kinetic energy is:
- (A) $KE = mgh$
 - (B) $KE = (1/2)mv^2$
 - (C) $KE = Fd$
 - (D) $KE = P/t$

18. The formula for gravitational potential energy is:
- (A) $PE = mgh$
 - (B) $PE = (1/2)mv^2$
 - (C) $PE = Fd$
 - (D) $PE = IV$
19. According to the law of conservation of energy, energy:
- (A) can be created but not destroyed
 - (B) can be destroyed but not created
 - (C) cannot be created or destroyed, only transformed
 - (D) is always lost as heat
20. For a wave travelling at constant speed, a higher frequency corresponds to a:
- (A) longer wavelength
 - (B) shorter wavelength
 - (C) identical wavelength
 - (D) wave with no wavelength
21. Which of the following is a vector quantity?
- (A) Speed
 - (B) Distance
 - (C) Mass
 - (D) Velocity
22. The efficiency of a machine is calculated as:
- (A) useful output energy / total input energy x 100%
 - (B) total input energy / useful output energy x 100%
 - (C) output energy minus input energy
 - (D) input energy plus output energy
23. The SI unit of frequency is the:
- (A) metre
 - (B) second
 - (C) hertz
 - (D) newton
24. Which statement best describes an electrical insulator?
- (A) It allows electric current to flow through it easily
 - (B) It does not allow electric current to flow through it easily
 - (C) It is always magnetic
 - (D) It is always transparent
25. Sound is best classified as a:
- (A) transverse wave
 - (B) longitudinal wave
 - (C) electromagnetic wave
 - (D) standing wave only
26. Which formula correctly relates speed, distance, and time?
- (A) speed = distance x time
 - (B) speed = distance / time
 - (C) speed = time / distance
 - (D) distance = speed + time

Circle T if the statement is true, or F if it is false.

27. T / F Current flows more easily through a conductor with low resistance.
28. T / F In a series circuit, the current is the same at every point in the circuit.
29. T / F A concave mirror always produces a virtual image, no matter where the object is placed.
30. T / F Electrical power can be calculated using $P = VI$ (voltage multiplied by current).
31. T / F Radiation requires a medium (such as air) in order to transfer heat energy.
32. T / F Adding more resistors in parallel decreases the total resistance of a circuit.
33. T / F The north pole of a magnet repels the south pole of another magnet.
34. T / F Light travels more slowly in glass than it does in air.
35. T / F A step-up transformer increases voltage while decreasing current (assuming no energy losses).
36. T / F Specific heat capacity is the energy needed to change a substance's state without changing its temperature.
37. T / F Electromagnetic induction can generate an electric current by moving a magnet near a coil of wire.
38. T / F All metals have exactly the same specific heat capacity as each other.
39. T / F A convex mirror always forms a virtual, upright, diminished image, regardless of object distance.
40. T / F Electrical energy transferred can be calculated using $E = Pt$, where P is power and t is time.
41. T / F Sound waves are able to travel through a vacuum.
42. T / F Total internal reflection can occur when light travels from a denser medium into a less dense medium at a large enough angle.
43. T / F Kinetic energy depends on both the mass and the velocity of a moving object.
44. T / F Gravitational potential energy depends on an object's height above a reference point.
45. T / F At this level, energy is described as something that cannot be created or destroyed, only transformed.
46. T / F Velocity and speed are exactly the same physical quantity.
47. T / F No real machine can be 100% efficient, because some energy is always lost, often as heat.
48. T / F For a wave travelling at constant speed, wavelength and frequency are directly proportional to each other.
49. T / F The SI unit of frequency is the hertz (Hz).
50. T / F An electrical insulator allows current to flow through it easily.
51. T / F Sound waves are longitudinal waves.
52. T / F Speed is calculated by dividing distance by time.

Fill in each blank with the correct word, phrase, or value.

53. The SI unit of electrical resistance is the ____.
54. A component that converts electrical energy into light energy is called a ____.
55. The transfer of heat through a fluid due to differences in density is called ____.
56. Electrical power can be written as $P = I^2 \times$ ____.

57. A lens that is thinner in the middle than at the edges is called a ____ lens.
58. The magnetic field lines around a straight current-carrying wire form concentric ____.
59. The total resistance of resistors connected in series is found by ____ the individual resistances.
60. The temperature at which a substance changes from a solid to a liquid is called its ____ point.
61. The heat energy needed to raise the temperature of 1 kg of a substance by 1 degree C is called its ____.
62. The energy needed to change a substance from solid to liquid without changing its temperature is called ____ heat.
63. A device that uses electromagnetic induction to generate electricity is called a(n) ____.
64. A transformer that decreases voltage is called a(n) ____ transformer.
65. The bending of light as it passes from one transparent medium into another is called ____.
66. The unit used on electricity bills to measure energy transferred is the kilowatt-____.
67. The formula relating energy, power, and time is $E = P \times$ ____.
68. A mirror that curves outward, like the back of a spoon, is called a ____ mirror.
69. The formula for kinetic energy is $KE = (1/2) \times m \times$ ____.
70. The formula for gravitational potential energy is $PE = m \times g \times$ ____.
71. The law of conservation of energy states that energy cannot be created or ____, only transformed.
72. A quantity that has both magnitude and direction is called a ____ quantity.
73. The efficiency of a machine is the ratio of useful output energy to ____ input energy, as a percentage.
74. For a wave travelling at constant speed, wavelength and frequency are ____ proportional to each other.
75. The SI unit of frequency is the ____.
76. A material that does not allow electric current to flow through it easily is called a(n) ____.
77. Sound waves are an example of a ____ wave.
78. The formula for speed is distance divided by ____.

79. Table 1 - Ohm's Law calculations ($V = IR$)

Voltage (V)	Current (A)	Resistance (ohm)
12	2	_____
_____	5	2
9	_____	3

80. Table 2 - Combining two resistors R1 and R2

R1 (ohm)	R2 (ohm)	Series total ($R1+R2$)	Parallel total ($R1R2/(R1+R2)$)
4	4	_____	_____
6	3	9	_____

10	_____	20	5
----	-------	----	---

81. Table 3 - Kinetic energy calculations ($KE = 0.5 \times m \times v^2$)

Mass (kg)	Velocity (m/s)	Kinetic energy (J)
2	3	_____
_____	5	50
1	_____	8

82. For each material below, put a tick in the column that applies.

Item	Conductor	Insulator
Copper wire		
Rubber glove		
Glass rod		
Aluminium foil		
Plastic ruler		
Iron nail		
Wooden block		
Silver spoon		

Answer in one to three complete sentences.

83. Calculate the current through a 6 ohm resistor connected to a 12 V battery.

84. Two resistors of 5 ohm and 20 ohm are connected in parallel. Calculate the total resistance.

85. State one similarity and one difference between conduction and convection.

86. A converging lens has a focal length of 10 cm. An object is placed 25 cm from the lens (more than twice the focal length). State whether the resulting image is real or virtual, and whether it is magnified or diminished.

87. Explain why birds can safely perch on a high-voltage power line without being electrocuted.

88. Calculate the energy transferred by a 2000 W kettle running for 3 minutes (180 seconds).

89. State two differences between the images produced by a convex mirror and a concave mirror.

91. Describe, step by step, how you would connect a battery, a switch, a bulb, and connecting wires to light the bulb in a simple series circuit. Then explain what would happen to the bulb's brightness if a second, identical bulb were added (a) in series and (b) in parallel.

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92. Explain the difference between real images and virtual images formed by lenses and mirrors, and give one real-world device that relies on each type of image.

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93. Explain how a simple generator produces an electric current, referring to electromagnetic induction, and describe one everyday application of generators.

[illegible]

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Biology

Answer all questions in the spaces provided.

Choose the letter of the correct answer for each question.

1. Which organelle is known as the "powerhouse of the cell"?
 - (A) Nucleus
 - (B) Mitochondria
 - (C) Ribosome
 - (D) Golgi apparatus
2. The genetic material of a eukaryotic cell is mainly found in the:
 - (A) Cytoplasm
 - (B) Nucleus
 - (C) Cell membrane
 - (D) Vacuole
3. In Mendelian genetics, an organism with two identical alleles for a trait is described as:
 - (A) Heterozygous
 - (B) Homozygous
 - (C) Hybrid
 - (D) Mutant
4. Which chamber of the heart pumps oxygenated blood out to the rest of the body?
 - (A) Right atrium
 - (B) Right ventricle
 - (C) Left atrium
 - (D) Left ventricle
5. Gas exchange in the lungs takes place in tiny air sacs called:
 - (A) Bronchi
 - (B) Trachea
 - (C) Alveoli
 - (D) Bronchioles
6. A group of different species living and interacting in the same area is called a:
 - (A) Population
 - (B) Community
 - (C) Ecosystem
 - (D) Biome
7. The process by which green plants make their own food using light energy is called:
 - (A) Respiration
 - (B) Photosynthesis
 - (C) Transpiration
 - (D) Fermentation
8. If both parents are heterozygous carriers (Aa) for a recessive trait, what fraction of their offspring is expected to show the recessive phenotype?
 - (A) $\frac{1}{4}$
 - (B) $\frac{1}{2}$
 - (C) $\frac{3}{4}$
 - (D) 0

9. The double-helix structure of DNA is famously credited to:
- (A) Gregor Mendel
 - (B) Charles Darwin
 - (C) Watson and Crick
 - (D) Louis Pasteur
10. Which nitrogenous base is found only in DNA, not in RNA?
- (A) Adenine
 - (B) Guanine
 - (C) Cytosine
 - (D) Thymine
11. The process by which organisms with favorable inherited traits are more likely to survive and reproduce is called:
- (A) Mutation
 - (B) Natural selection
 - (C) Genetic drift
 - (D) Cloning
12. Which blood vessel carries deoxygenated blood away from the heart?
- (A) Aorta
 - (B) Pulmonary vein
 - (C) Pulmonary artery
 - (D) Vena cava
13. The type of cell division that produces gametes (sex cells) with half the normal number of chromosomes is called:
- (A) Mitosis
 - (B) Meiosis
 - (C) Binary fission
 - (D) Budding
14. In a food chain, organisms that make their own food using light energy are called:
- (A) Consumers
 - (B) Decomposers
 - (C) Producers
 - (D) Predators
15. Which of the following is an example of a decomposer?
- (A) Lion
 - (B) Oak tree
 - (C) Mushroom (fungus)
 - (D) Eagle
16. A person with blood type O is generally considered a universal:
- (A) Donor
 - (B) Recipient
 - (C) Carrier
 - (D) Mutant

17. Compared with the air breathed in, the air breathed out by humans contains more:
- (A) Oxygen
 - (B) Nitrogen
 - (C) Carbon dioxide
 - (D) Hydrogen
18. Which structure in a plant cell is mainly responsible for photosynthesis?
- (A) Mitochondria
 - (B) Chloroplast
 - (C) Nucleus
 - (D) Vacuole
19. The movement of water across a semi-permeable membrane from high to low water concentration is called:
- (A) Diffusion
 - (B) Osmosis
 - (C) Active transport
 - (D) Transpiration
20. A characteristic that is controlled by genes and passed from parent to offspring is called a(n):
- (A) Adaptation
 - (B) Inherited trait
 - (C) Acquired trait
 - (D) Mutation event
21. Which organ is mainly responsible for filtering waste products out of the blood?
- (A) Liver
 - (B) Kidney
 - (C) Heart
 - (D) Lung
22. The scientific (binomial) name of an organism is typically written using:
- (A) One word
 - (B) Two words (genus and species)
 - (C) Three words
 - (D) A numeric code
23. The process by which plants lose water vapor through small pores called stomata is called:
- (A) photosynthesis
 - (B) respiration
 - (C) transpiration
 - (D) germination
24. Which blood cell type is primarily responsible for fighting infection?
- (A) Red blood cells
 - (B) White blood cells
 - (C) Platelets
 - (D) Plasma
25. Which is the most specific (narrowest) level of biological classification?
- (A) Kingdom
 - (B) Genus
 - (C) Species
 - (D) Family

26. Which of these is NOT a function of the skeletal system?

- (A) Support
- (B) Protection of organs
- (C) Production of digestive enzymes
- (D) Movement (with muscles)

Circle T if the statement is true, or F if it is false.

27. T / F Ribosomes are the site of protein synthesis in a cell.

28. T / F Mature human red blood cells contain a nucleus.

29. T / F Dominant alleles are always more common in a population than recessive alleles.

30. T / F Blood is carried away from the heart through arteries.

31. T / F Photosynthesis occurs in the mitochondria of plant cells.

32. T / F An ecosystem includes both the living organisms and the non-living environment in an area.

33. T / F Oxygen diffuses from the alveoli into the blood because its concentration is higher in the alveoli than in the blood.

34. T / F All mutations are harmful to the organism that carries them.

35. T / F DNA is composed of a double helix made of two strands of nucleotides.

36. T / F Meiosis produces four genetically identical daughter cells.

37. T / F Natural selection acts on genetic variation that already exists within a population.

38. T / F Decomposers break down dead organic matter and recycle nutrients back into an ecosystem.

39. T / F All bacteria are harmful to humans.

40. T / F Charles Darwin proposed the theory of evolution by natural selection.

41. T / F RNA contains the base thymine instead of uracil.

42. T / F Mitosis results in two daughter cells that are genetically identical to the parent cell.

43. T / F Osmosis is the movement of water across a semi-permeable membrane from high to low water concentration.

44. T / F Chloroplasts are found in animal cells as well as in plant cells.

45. T / F Acquired traits, such as a scar or a learned skill, can be passed on genetically to offspring.

46. T / F The kidneys play a major role in filtering waste products from the blood and producing urine.

47. T / F Binomial nomenclature names an organism using its genus and species.

48. T / F All living things are made up of at least one cell.

49. T / F Transpiration is the loss of water vapor from a plant's leaves through the stomata.

50. T / F White blood cells are primarily responsible for helping the body fight infection.

51. T / F Species is a broader classification category than kingdom.

52. T / F The skeletal system's main functions include producing digestive enzymes.

Fill in each blank with the correct word, phrase, or value.

53. The basic unit of heredity, passed from parent to offspring, is called a(n) ____.

54. The cell structure that controls what substances enter and leave the cell is the ____.
55. An organism's observable physical characteristics are called its ____, while its genetic makeup is called its ____.
56. The muscular organ that pumps blood around the body is the ____.
57. Gas and nutrient exchange between blood and body tissues occurs across thin-walled vessels called ____.
58. A stable community of organisms together with their physical environment is called a(n) ____.
59. Plants absorb the gas ____ from the air during photosynthesis.
60. For any gene, an offspring receives one allele from ____ parent.
61. The molecule that carries genetic instructions in living organisms is called ____.
62. The type of cell division that produces two genetically identical daughter cells for growth and repair is called ____.
63. Organisms that break down dead organic material and recycle nutrients are called ____.
64. Darwin's theory explaining how species change over time through survival of the fittest is called ____.
65. A change in an organism's DNA sequence is called a ____.
66. The four nitrogenous bases found in DNA are adenine, thymine, cytosine, and ____.
67. The natural environment in which an organism lives is called its ____.
68. The scientific practice of grouping living things based on shared characteristics is called ____.
69. The movement of water across a semi-permeable membrane from high to low water concentration is called ____.
70. The organelle mainly responsible for photosynthesis in plant cells is the ____.
71. The organs that filter waste products from the blood and produce urine are the ____.
72. A trait passed from parent to offspring through genes, rather than gained during life, is called a(n) ____ trait.
73. The naming system that gives every organism a two-part Latin name is called ____ nomenclature.
74. The basic structural and functional unit of all living organisms is the ____.
75. The loss of water vapor from a plant's leaves through pores called stomata is called ____.
76. The type of blood cell mainly responsible for fighting infection is the ____ blood cell.
77. The most specific (narrowest) level of biological classification is the ____.
78. Bones provide support and protection, and enable ____ when combined with muscles.

79. Table 1 - Punnett square for a cross between two heterozygous parents (Aa x Aa)

	A	a
A	AA	_____
a	_____	aa

80. Table 2 - Organs and their main functions

Organ	Main function	Body system
Heart	Pumps blood around the body	_____
Lungs	_____	Respiratory
_____	Filters blood and removes waste	Excretory

81. Table 3 - Cell transport processes

Process	Requires energy (ATP)?	Direction of movement
Diffusion	_____	High to low concentration
Osmosis (water)	No	_____
Active transport	Yes	_____

82. For each structure below, put a tick in the column that best describes where it is found.

Item	Plant cells only	Animal cells only	Both plant and animal cells
Cell wall			
Mitochondria			
Chloroplast			
Nucleus			
Large central vacuole			
Ribosomes			
Cell membrane			
Centrioles			

Answer in one to three complete sentences.

83. State two differences between arteries and veins.

84. In pea plants, purple flower color (P) is dominant over white (p). A heterozygous plant (Pp) is crossed with a white-flowered plant (pp). Show the cross and state what fraction of the offspring is expected to have purple flowers.

85. Explain why alveoli are well adapted for efficient gas exchange (state and explain two features).

86. Define "population" and "community" in ecological terms, and explain the difference between them.

87. Explain what is meant by the term "homozygous recessive" and give an example genotype.

88. State one similarity and one difference between mitosis and meiosis.

89. Explain, in one or two sentences, how antibiotic resistance in bacteria can develop through natural selection.

90. Name the four nitrogenous bases found in DNA, and state which two always pair together.

Write a well-organized, extended response of about 120-200 words.

91. Describe the pathway of blood through the human heart and lungs in one complete circuit, starting from the right atrium, and explain why this double circulation system helps deliver oxygen efficiently to body tissues.

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English

Answer all questions in the spaces provided.

Reading Passage

The Rise of Renewable Energy in Southeast Asia

Over the past decade, renewable energy has grown faster in Southeast Asia than any other power source. Solar power, in particular, has expanded rapidly across countries such as Thailand, Vietnam, and the Philippines. Two main factors have driven this growth. First, the cost of manufacturing solar panels has fallen sharply, making solar electricity cheaper than ever before. Second, many governments in the region have introduced incentives, such as tax breaks and subsidies, to encourage both households and businesses to install solar systems.

Wind power has also seen significant growth, especially in coastal areas where strong, steady winds make large turbines practical. Engineers have installed new wind farms along several coastlines, adding thousands of megawatts of clean electricity to national grids.

Despite this progress, fossil fuels such as coal and natural gas still supply the majority of the region's electricity. Experts argue that continued investment in renewable technology, along with supportive government policy, will be necessary if Southeast Asian countries are to meet their long-term climate targets.

Choose the letter of the correct answer for each question.

1. Choose the correct verb form: "By next year, she ____ her studies."
(A) will have completed
(B) completing
(C) completes
(D) has completed
2. Which word is closest in meaning to "abundant"?
(A) Scarce
(B) Plentiful
(C) Expensive
(D) Fragile
3. Which sentence is correctly punctuated?
(A) Its a beautiful day, isnt it
(B) It's a beautiful day, isn't it?
(C) Its' a beautiful day, isnt' it?
(D) It is a beautiful day isnt it
4. Which sentence uses the passive voice correctly?
(A) The cake was eaten by the children.
(B) The children eating the cake.
(C) The cake eat the children.
(D) Eaten the cake children.

5. According to the passage, what has helped drive the growth of solar power in the region?
- (A) Rising panel costs and fewer regulations
 - (B) Falling panel costs and government incentives
 - (C) A shortage of fossil fuels
 - (D) A decrease in electricity demand
6. Which word is an antonym (opposite) of "increase"?
- (A) Rise
 - (B) Expand
 - (C) Decrease
 - (D) Grow
7. Which of the following is the correctly spelled word?
- (A) Recieve
 - (B) Receive
 - (C) Recievee
 - (D) Receve
8. Choose the conjunction that best completes the sentence: "She studied hard, ____ she passed the exam."
- (A) but
 - (B) so
 - (C) or
 - (D) yet
9. Choose the correctly formed plural of "child":
- (A) childs
 - (B) children
 - (C) childes
 - (D) childrens
10. Which sentence is written in the correct verb tense?
- (A) Yesterday, I go to the market.
 - (B) Yesterday, I went to the market.
 - (C) Yesterday, I going to the market.
 - (D) Yesterday, I goes to the market.
11. Which word is a synonym for "happy"?
- (A) miserable
 - (B) joyful
 - (C) angry
 - (D) tired
12. Choose the correct preposition: "The book is ____ the table."
- (A) on
 - (B) in
 - (C) at
 - (D) by
13. Which sentence contains a simile?
- (A) The moon is a bright lamp in the sky.
 - (B) Her smile was as bright as the sun.
 - (C) The sun smiled down on us.
 - (D) The wind whispered through the trees.

14. Which sentence is capitalized correctly?
- (A) i live in bangkok, thailand.
 - (B) I live in Bangkok, Thailand.
 - (C) I live in bangkok, Thailand.
 - (D) I Live In Bangkok, Thailand.
15. Which of these words is spelled correctly?
- (A) definately
 - (B) definitely
 - (C) definitly
 - (D) deffinitely
16. Which sentence is a correctly formed question?
- (A) Where you are going?
 - (B) Where are you going?
 - (C) Where you going are?
 - (D) You are going where?
17. Which sentence uses correct subject-verb agreement?
- (A) The list of items are on the desk.
 - (B) The list of items is on the desk.
 - (C) The list of items be on the desk.
 - (D) The list of items being on the desk.
18. Which sentence uses the correct homophone?
- (A) Their going to the park.
 - (B) They're going to the park.
 - (C) There going to the park.
 - (D) Theyre going to the park.
19. Which sentence is an example of alliteration?
- (A) The big brown bear bounced happily.
 - (B) The dog ran across the yard.
 - (C) It was a dark and stormy night.
 - (D) The sun rose over the mountains.
20. Choose the correct verb form: "Neither of the answers ____ correct."
- (A) are
 - (B) is
 - (C) were
 - (D) be
21. In the sentence "I wanted to go, but it was raining," which word is the conjunction?
- (A) wanted
 - (B) but
 - (C) raining
 - (D) go
22. Which sentence contains an example of onomatopoeia?
- (A) The bees buzzed around the flowers.
 - (B) The sky was blue.
 - (C) She walked to school.
 - (D) He read a book.

23. Which sentence correctly uses an apostrophe to show singular possession?

- (A) The dogs bone.
- (B) The dog's bone.
- (C) The dogs' bone.
- (D) The doges bone.

24. Choose the correct comparative form: "This book is _____ than that one."

- (A) more interesting
- (B) interestinger
- (C) most interesting
- (D) interesting

25. Which of these words is a collective noun?

- (A) team
- (B) player
- (C) game
- (D) score

26. Which sentence is written in the active voice?

- (A) The ball was thrown by Tom.
- (B) Tom threw the ball.
- (C) The ball is thrown.
- (D) Thrown was the ball.

Circle T if the statement is true, or F if it is false.

27. T / F According to the passage, the cost of solar energy has risen sharply over the past decade.

28. T / F A noun is a word that describes an action.

29. T / F According to the passage, government incentives have supported the growth of renewable energy in the region.

30. T / F An adverb can modify a verb, an adjective, or another adverb.

31. T / F According to the passage, fossil fuels are described as the fastest-growing energy source in Southeast Asia.

32. T / F The sentence "Running quickly, the bus was caught by him" contains a dangling modifier.

33. T / F According to the passage, wind power has seen no growth in the region.

34. T / F A comma splice occurs when two independent clauses are joined only by a comma.

35. T / F "Children" is the correct plural form of "child."

36. T / F A simile makes a comparison using the words "like" or "as."

37. T / F A metaphor makes a direct comparison without using "like" or "as."

38. T / F The sentence "Me and him went to the store" is grammatically correct standard English.

39. T / F An adjective describes or modifies a noun.

40. T / F The past tense of "go" is "goed."

41. T / F Personification gives human qualities to non-human things or objects.

42. T / F A synonym is a word that means the opposite of another word.

43. T / F "They're," "their," and "there" are homophones that are spelled and used identically.

44. T / F Alliteration is the repetition of the same initial consonant sound in nearby words.
45. T / F The word "neither" is treated as a singular subject in formal English grammar.
46. T / F Onomatopoeia refers to words that imitate the sound they describe, such as "buzz" or "splash."
47. T / F A run-on sentence correctly joins two independent clauses without proper punctuation or conjunctions.
48. T / F The subject of an English sentence must always come before the verb.
49. T / F An apostrophe followed by "s" is used to show singular possession, as in "the dog's bone."
50. T / F The comparative form of "interesting" is "interestinger."
51. T / F A collective noun refers to a group treated as a single unit, such as "team" or "flock."
52. T / F The sentence "Tom threw the ball" is written in the passive voice.

Fill in each blank with the correct word, phrase, or value.

53. She ____ (finish) her homework before dinner yesterday.
54. Neither the teacher nor the students ____ (be) ready for the fire drill.
55. This is the book ____ I told you about.
56. If I ____ (be) you, I would apologize.
57. The opposite of "ancient" is ____.
58. A word that joins two clauses or sentences together is called a(n) ____.
59. The comparative form of the adjective "good" is ____.
60. He is one of the students who ____ (have) finished the project.
61. The plural of "mouse" (the animal) is ____.
62. A word that describes a noun is called a(n) ____.
63. A comparison using the words "like" or "as" is called a ____.
64. The past tense of the verb "write" is ____.
65. A word that means the same, or nearly the same, as another word is called a ____.
66. In "He quickly ran to the store," the word "quickly" is a(n) ____.
67. The subject of the sentence "The dog barked loudly" is ____.
68. A group of words that expresses a complete thought is called a ____.
69. The repetition of the same initial consonant sound in a phrase, such as "big brown bear," is called ____.
70. A word that imitates the sound it describes, such as "buzz" or "splash," is called a(n) ____.
71. Two independent clauses joined without proper punctuation or a conjunction form a(n) ____ sentence.
72. The words "their," "there," and "they're" are examples of ____ - words that sound alike but differ in meaning and spelling.
73. In formal grammar, the pronoun "neither" takes a ____ verb.
74. A sentence fragment is a group of words that does not express a ____ thought.

75. An apostrophe followed by "s," as in "the dog's bone," shows singular ____.

76. The comparative form of "interesting" is "____ interesting."

77. A noun referring to a group treated as a single unit, such as "team" or "flock," is called a ____ noun.

78. In "Tom threw the ball," the sentence is written in the ____ voice.

79. Table 1 - Identify the part of speech of the bracketed word in each sentence

Sentence	Word	Part of speech
She sings (beautifully).	beautifully	_____
The (quick) fox jumped.	quick	_____
He (runs) every morning.	runs	_____
(Happiness) is contagious.	Happiness	_____

80. Table 2 - Verb tense forms

Base form	Past simple	Past participle
go	_____	gone
eat	ate	_____
begin	_____	begun
write	wrote	_____

81. Table 3 - Choose the correct homophone to complete each sentence

Sentence	Word choices	Correct word
I can't find ____ keys.	their / there / they're	_____
____ going to be late.	their / there / they're	_____
Put the book over ____.	their / there / they're	_____

82. For each word below, put a tick in the column showing its part of speech.

Item	Noun	Verb	Adjective
happiness			
run			
beautiful			
honesty			
jump			
curious			
kindness			

swim			
difficult			

Answer in one to three complete sentences.

83. According to the passage, name two reasons for the growth of solar power mentioned in the text.

84. Rewrite the following sentence in the passive voice: "The engineers installed the new solar panels."

85. Combine the following two sentences using an appropriate conjunction: "It was raining. We went for a walk."

86. Identify and correct the grammatical error in this sentence: "Each of the students have submitted their assignment."

87. In your own words, summarize the main idea of the passage in one or two sentences.

88. Write one sentence that uses a simile to describe a thunderstorm.

89. Correct the grammar error in this sentence: "Me and my friend is going to the cinema."

90. Identify the adjectives in this sentence and the noun each one describes: "The curious cat explored the dark attic."

Write a well-organized, extended response of about 120-200 words.

91. Write a short paragraph (about 120-150 words) describing the advantages and disadvantages of renewable energy, based on the passage and your own knowledge.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

92. Write a short persuasive letter (150-200 words) to your school principal proposing one environmental change your school should make, including at least two supporting reasons.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

93. Write a short descriptive paragraph (about 100-150 words) about a place that is special to you, using at least one simile or metaphor.

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94. Write a short narrative paragraph (about 100-150 words) that includes at least one example of alliteration and one example of onomatopoeia. Underline or clearly mark both examples.

[illegible]